

# **QUESTION 1**

## Mathematical Calculator

```
while True:
    print("\n===== MATHEMATICAL CALCULATOR =====")
    print("1. Addition (+)")
    print("2. Subtraction (-)")
    print("3. Multiplication (*)")
    print("4. Division (/)")
    print("5. Floor Division (//)")
    print("6. Power (^)")
    print("7. Modulus (%)")
    print("8. Clear")
    print("9. OFF")

    choice = input("Enter your choice: ")

    if choice == "9":
        print("Calculator OFF.")
        break

    elif choice == "8":
        print("\n" * 50)
        continue

    elif choice in ["1", "2", "3", "4", "5", "6", "7"]:

        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        if choice == "1":
            print("Answer =", num1 + num2)

        elif choice == "2":
            print("Answer =", num1 - num2)

        elif choice == "3":
            print("Answer =", num1 * num2)

        elif choice == "4":
            if num2 == 0:
                print("Cannot divide by zero.")
```

```

else:
    print("Answer =", num1 / num2)

elif choice == "5":
    if num2 == 0:
        print("Cannot divide by zero.")
    else:
        print("Answer =", num1 // num2)

elif choice == "6":
    print("Answer =", num1 ** num2)

elif choice == "7":
    if num2 == 0:
        print("Cannot divide by zero.")
    else:
        print("Answer =", num1 % num2)

else:
    print("Invalid choice.")

```

## **QUESTION 2**

Personal Pocket CGPA Calculator

```

print("===== PERSONAL POCKET CGPA CALCULATOR =====")

courses = int(input("Enter number of courses: "))

total_points = 0
total_units = 0

for i in range(courses):
    print("\nCourse", i + 1)

    unit = int(input("Course Unit: "))
    score = int(input("Course Score: "))

    if score >= 70:
        gp = 5
        grade = "A"

```

```
elif score >= 60:  
    gp = 4  
    grade = "B"
```

```
elif score >= 50:  
    gp = 3  
    grade = "C"
```

```
elif score >= 45:  
    gp = 2  
    grade = "D"
```

```
elif score >= 40:  
    gp = 1  
    grade = "E"
```

```
else:  
    gp = 0  
    grade = "F"
```

```
print("Grade:", grade)  
print("Grade Point:", gp)
```

```
total_points += gp * unit  
total_units += unit
```

```
cgpa = total_points / total_units
```

```
print("\n=====")  
print("Total Units:", total_units)  
print("Total Grade Points:", total_points)  
print("CGPA =", round(cgpa, 2))  
print("=====")
```